SUMS OF RANDOM VARIABLES

Lect. İsmail BAŞOĞLU, Ph.D.

Özyeğin University

February 2, 2024

1 Expected Value of Sums of Random Variables
- Expected Value of the Sum of Two Random Variables
- Expected Value of the Sum of n Random Variables
- Expected Value of Sums of RVs

1. Expected Value of Sums of Random Variables
   - Expected Value of the Sum of Two Random Variables
   - Expected Value of the Sum of n Random Variables
   - Expected Value of Sums of RVs
2. Covariance & Correlation
   - Definition of Covariance
   - Further Properties of Covariance
   - Definition of Correlation
   - Further Properties of Correlation

1. Expected Value of Sums of Random Variables
   - Expected Value of the Sum of Two Random Variables
   - Expected Value of the Sum of n Random Variables
   - Expected Value of Sums of RVs
2. Covariance & Correlation
   - Definition of Covariance
   - Further Properties of Covariance
   - Definition of Correlation
   - Further Properties of Correlation
3. Variance of Sums of Random Variables
   - Variance of the Sum of Two Random Variables
   - Variance of Sums of n Random Variables

1. Expected Value of Sums of Random Variables
   - Expected Value of the Sum of Two Random Variables
   - Expected Value of the Sum of n Random Variables
   - Expected Value of Sums of RVs
2. Covariance & Correlation
   - Definition of Covariance
   - Further Properties of Covariance
   - Definition of Correlation
   - Further Properties of Correlation
3. Variance of Sums of Random Variables
   - Variance of the Sum of Two Random Variables
   - Variance of Sums of n Random Variables
4. Additive Relationships Between Special Random Variables
   - Sum of iid Bernoulli RVs is A Binomial RV
   - Sum of Independent Poisson RVs is A Poisson RV
   - Sum of iid Geometric RVs is A Negative Binomial RV
   - Sum of Normal RVs is A Normal RV

Expected Value of the Sum of Two Random Variables

- **Theorem:** Suppose we are given two random variables X and Y. The expected value of the sum of these random variables is equal to the sum of their expectations. Namely:

$$E[X+Y] = E[X] + E[Y]$$

Expected Value of the Sum of Two RVs (cont'd)

- **Proof:** Suppose both X and Y are continuous[1] RVs with joint PDF $f_{XY}(x,y)$.

$$
\begin{aligned}
E[X+Y] &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x+y)\,f_{XY}(x,y)\,dxdy \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf_{XY}(x,y)+yf_{XY}(x,y)\,dxdy \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf_{XY}(x,y)\,dxdy+\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} yf_{XY}(x,y)\,dxdy \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf_{XY}(x,y)\,dydx+\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} yf_{XY}(x,y)\,dxdy \\
&= \int_{-\infty}^{\infty} x\int_{-\infty}^{\infty} f_{XY}(x,y)\,dydx+\int_{-\infty}^{\infty} y\int_{-\infty}^{\infty} f_{XY}(x,y)\,dxdy \\
&= \int_{-\infty}^{\infty} xf_X(x)dx+\int_{-\infty}^{\infty} yf_Y(y)dy \\
&= E[X]+E[Y]
\end{aligned}
$$

[1] The proof is similar for other cases

Expected Value of the Sum of n Random Variables

- **Proposition:** Say X_i, $i = 1, \ldots, n$, is a sequence of random variables. The expected value of their sum is equal to the sum of their expectations.

$$E\left[X_1 + \ldots + X_n\right] = E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E\left[X_i\right] = E\left[X_1\right] + \ldots + E\left[X_n\right]$$

Further Properties of Expected Value of Sums of RVs

- Let $a, b \in \mathbb{R}$ be two arbitrary real numbers, and X and Y be two random variables.

$$E[aX + bY] = aE[X] + bE[Y]$$

- Let g and h be two real-valued functions, and X and Y be two random variables. Since a function of any random variable is another random variable:

$$E[g(X) + h(Y)] = E[g(X)] + E[h(Y)]$$

Expected Value of Sums of RVs: Example

Q1. Let X and Y be two random variables. We know that $E[X] = 5$, $Var[X] = 9$, $E[Y] = 4$, $Var[Y] = 4$, and $E[XY] = 15$.

a. Calculate $E\left[(X+2Y)^2\right]$,
b. Calculate $E[-2X+3Y]$,
c. Calculate $E\left[-2X^2+(2Y)^2\right]$,
d. Calculate $E\left[X^2+Y^3\right]$,

if it is possible with the provided information.

A1. Along with the provided information, we can also derive:

$$E\left[X^2\right] = Var[X] + E[X]^2 = 9 + 5^2 = 34$$
$$E\left[Y^2\right] = Var[Y] + E[Y]^2 = 4 + 4^2 = 20$$

Expected Value of Sums of RVs: Example (cont'd)

a. The square of the sum can be expanded:

$$\begin{aligned} E\left[(X+2Y)^2\right] &= E\left[X^2+4XY+4Y^2\right] \\ &= E\left[X^2\right]+4E\left[XY\right]+4E\left[Y^2\right] \\ &= 34+4\times 15+4\times 20 = 174 \end{aligned}$$

b. $E\left[-2X+3Y\right] = -2E\left[X\right]+3E\left[Y\right] = -2\times 5+3\times 4 = 2$

c. The square can be expanded:

$$\begin{aligned} E\left[-2X^2+(2Y)^2\right] &= E\left[-2X^2+4Y^2\right] \\ &= -2E\left[X^2\right]+4E\left[Y^2\right] \\ &= -2\times 34+4\times 20 = 12 \end{aligned}$$

d. $E\left[X^2+Y^3\right]$ can not be computed since $E\left[Y^3\right]$ can not be derived with the provided information.

What is Covariance?

- Expected value and variance gives us partial information about a single random variable.
  - Expected value gives us information on the **central tendency** of a random variable.
    - **Expected value** of a RV is also the mean value of that RV.
  - **Variance** gives us information on the **dispersion (spread)** of a random variable.
- **Covariance** is defined between two random variables and gives us information about the **linear dependence** relationship between the random variables.

Definition of Covariance

- Suppose we are given two random variables X and Y. The **covariance** of X and Y is defined as:

$$Cov[X,Y] = E[(X - E[X])(Y - E[Y])]$$

- The covariance formula can be transformed to following form:

$$\begin{aligned} Cov[X,Y] &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY - XE[Y] - YE[X] + E[X]E[Y]] \\ &= E[XY] - E[XE[Y]] - E[YE[X]] + E[E[X]E[Y]] \\ &= E[XY] - E[X]E[Y] - E[X]E[Y] + E[X]E[Y] \\ &= E[XY] - E[X]E[Y] \end{aligned}$$

Definition of Covariance (cont'd)

- Hence, to calculate the covariance of X and Y, there is a much simpler equation, that is:

$$Cov[X,Y] = E[XY] - E[X]E[Y]$$

- Note that, unlike the variance, covariance can take negative values as well.
  - When $Cov[X,Y] > 0$, there is a positive linear dependence relationship between X and Y. In other words, **when X takes high values, Y tends to take high values as well.**
  - When $Cov[X,Y] < 0$, there is a negative linear dependence relationship between X and Y. In other words, **when X takes high values, Y tends to take low values.**

Covariance of Independent RVs

- **Proposition:** When X and Y are independent, $Cov[X,Y]=0$.
- **Proof:** Suppose both X and Y are continuous[2] RVs with joint PDF $f_{XY}(x,y)$. When X and Y are independent, we have:

$$f_{XY}(x,y)=f_X(x)f_Y(y), \quad \forall x,y\in\mathbb{R}$$

When we use this in $E[XY]$:

$$\begin{aligned}E[XY] &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xyf_{XY}(x,y)\,dxdy = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xyf_X(x)f_Y(y)\,dxdy \\ &= \int_{-\infty}^{\infty} yf_Y(y)\int_{-\infty}^{\infty} xf_X(x)\,dxdy = \int_{-\infty}^{\infty} yf_Y(y)E[X]\,dy \\ &= E[X]\int_{-\infty}^{\infty} yf_Y(y)\,dy = E[X]E[Y]\end{aligned}$$

Hence $Cov[X,Y]=E[XY]-E[X]E[Y]=0$.

[2] The proof is similar for other cases

Further Properties of Covariance

- The fact that $Cov[X,Y]=0$ does not necessarily make X and Y independent.
  - There still can be a **nonlinear dependence** between X and Y.
  - Still there may be some $x,y \in \mathbb{R}$ such that:

$$F_X(x)F_Y(y) \neq F_{XY}(x,y)$$

- Since covariance is a symmetric relationship:

$$Cov[X,Y]=Cov[Y,X]$$

- Covariance of a random variable X with itself is equal to its variance, since:

$$Cov[X,X]=E\left[X^2\right]-E[X]E[X]=E\left[X^2\right]-E[X]^2=Var[X]$$

Further Properties of Covariance (cont'd)

- **Proposition:** Let a be any real number. Then:

$$Cov[aX,Y] = aCov[X,Y]$$

- **Proof:**

$$\begin{aligned} Cov[aX,Y] &= E[aXY] - E[aX]E[Y] \\ &= aE[XY] - aE[X]E[Y] \\ &= aCov[X,Y] \end{aligned}$$

- Similarly, if we let $a,b \in \mathbb{R}$. Then:

$$Cov[aX,bY] = abCov[X,Y]$$

Further Properties of Covariance (cont'd)

- **Proposition:** Let X_i, $i = 1, \ldots, m$ and Y_j, $j = 1, \ldots, n$ be two sequences of random variables. Then:

$$Cov\left[\sum_{i=1}^{m} X_i, \sum_{j=1}^{n} Y_j\right] = \sum_{i=1}^{m}\sum_{j=1}^{n} Cov\left[X_i, Y_j\right]$$

Namely, the covariance of the sums of two sequences, is the sum of the covariances of all possible dual combinations.

Further Properties of Covariance (cont'd)

- **Proof:**

$$
\begin{aligned}
Cov\left[\sum_{i=1}^{m} X_i, \sum_{j=1}^{n} Y_j\right] &= E\left[\left(\sum_{i=1}^{m} X_i - E\left[\sum_{i=1}^{m} X_i\right]\right)\left(\sum_{j=1}^{n} Y_j - E\left[\sum_{i=1}^{n} Y_j\right]\right)\right] \\
&= E\left[\left(\sum_{i=1}^{m} X_i - \sum_{i=1}^{m} E\left[X_i\right]\right)\left(\sum_{j=1}^{n} Y_j - \sum_{j=1}^{n} E\left[Y_j\right]\right)\right] \\
&= E\left[\left(\sum_{i=1}^{m} \left(X_i - E\left[X_i\right]\right)\right)\left(\sum_{j=1}^{n} \left(Y_j - E\left[Y_j\right]\right)\right)\right] \\
&= E\left[\sum_{i=1}^{m}\sum_{j=1}^{n} \left(X_i - E\left[X_i\right]\right)\left(Y_j - E\left[Y_j\right]\right)\right] \\
&= \sum_{i=1}^{m}\sum_{j=1}^{n} E\left[\left(X_i - E\left[X_i\right]\right)\left(Y_j - E\left[Y_j\right]\right)\right] = \sum_{i=1}^{m}\sum_{j=1}^{n} Cov\left[X_i, Y_j\right]
\end{aligned}
$$

Covariance: Example

Q2. Suppose that X and Y have the following joint PMF:

$p_{XY}(x,y)$	y		
x	0	2	4
1	0.20	0.20	0.10
2	0.10	0.05	0.10
3	0.10	0.05	0.10

a. Calculate $Cov[X,Y]$.
b. Calculate $Cov[3X,-2Y]$.

A2. We will compute each component in the following formula:

$$Cov[X,Y] = E[XY] - E[X]E[Y]$$

Covariance: Example (cont'd)

A2. Remind yourself with the marginal distributions of X and Y.

$p_{XY}(x,y)$	y		
x	0	2	4
1	0.20	0.20	0.10
2	0.10	0.05	0.10
3	0.10	0.05	0.10

$$p_X(x) = \begin{cases} 0.50, & x = 1 \\ 0.25, & x = 2 \\ 0.25, & x = 3 \\ 0, & \text{e/w} \end{cases} \qquad p_Y(y) = \begin{cases} 0.40, & y = 0 \\ 0.30, & y = 2 \\ 0.30, & y = 4 \\ 0, & \text{e/w} \end{cases}$$

$$E[X] = 1.75, \qquad E[Y] = 1.8$$

Covariance: Example (cont'd)

A2. Now let us compute $E[XY]$.

$p_{XY}(x,y)$	y		
x	0	2	4
1	0.20	0.20	0.10
2	0.10	0.05	0.10
3	0.10	0.05	0.10

$$\begin{aligned} E[XY] &= \sum_{i=1}^{3} \sum_{j\in\{0,2,4\}} ij\,p_{XY}(i,j) \\ &= 0\times 0.20 + 2\times 0.20 + 4\times 0.10 \\ &+ 0\times 0.10 + 4\times 0.05 + 8\times 0.10 \\ &+ 0\times 0.10 + 6\times 0.05 + 12\times 0.10 \\ &= 0.80 + 1 + 1.50 = 3.30 \end{aligned}$$

a. $Cov[X,Y] = E[XY] - E[X]E[Y] = 3.30 - (1.75)(1.8) = 0.15$

b. $Cov[3X,-2Y] = (3)(-2)Cov[X,Y] = -0.90$

Covariance: Example II

Q3. Suppose the joint PDF of X and Y is given by:

$$f_{XY}(x,y) = \begin{cases} 2-x-y, & 0 \leq x \leq 1,\ 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find $Cov[X,Y]$

Covariance: Example II

Q3. Suppose the joint PDF of X and Y is given by:

$$f_{XY}(x,y) = \begin{cases} 2-x-y, & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find $Cov[X,Y]$

A3. Remind yourself with the marginal distributions of X and Y.

$$f_X(x) = \frac{3}{2} - x, \quad x \in [0,1]$$

$$E[X] = \int_0^1 x f_X(x)\,dx = \int_0^1 \left(\frac{3x}{2} - x^2\right) dx = \left[\frac{3x^2}{4} - \frac{x^3}{3}\right]_{x=0}^{x=1} = \frac{5}{12}$$

Notice that $f_Y(y) = \frac{3}{2} - y$ for $y \in [0,1]$ and $E[Y] = \frac{5}{12}$ as well.

Covariance: Example II (cont'd)

A3. Now let us compute $E[XY]$.

$$\begin{aligned}
E[XY] &= \int_0^1 \int_0^1 xy f_{XY}(x,y)\,dxdy \\
&= \int_0^1 \int_0^1 \left(2xy - x^2y - xy^2\right) dxdy \\
&= \int_0^1 \left[x^2y - \frac{x^3y}{3} - \frac{x^2y^2}{2}\right]_0^1 dy \\
&= \int_0^1 \frac{2y}{3} - \frac{y^2}{2} dy \\
&= \left[\frac{y^2}{3} - \frac{y^3}{6}\right]_0^1 = \frac{1}{6}
\end{aligned}$$

Hence:

$$Cov[X,Y] = E[XY] - E[X]E[Y] = \frac{1}{6} - \left(\frac{5}{12}\right)\left(\frac{5}{12}\right) = -\frac{1}{144}$$

Definition of Correlation

- Just like covariance, correlation gives information on how two random variables are related.
- It is a better measure, than covariance, in indicating the **linear relationship** between the variables.
- Let X and Y be two random variables. **Correlation** of X and Y is defined as:

$$Corr[X,Y] = \frac{Cov[X,Y]}{\sqrt{Var[X]Var[Y]}}$$

- Hence, in order to calculate $Corr[X,Y]$, we need to know the covariance between X and Y and their respective variances.

Further Properties on Correlation

- Just like the covariance, correlation is also a symmetric relationship:

$$Corr[X,Y] = Corr[Y,X]$$

- Correlation between any two random variables may take values between -1 and 1.

$$-1 \leq Corr[X,Y] \leq 1$$

- If X and Y are independent, $Cov[X,Y] = 0$, therefore:

$$Corr[X,Y] = 0$$

Further Properties of Correlation (cont'd)

- Correlation of a random variable X with itself is equal to 1, since:

$$Corr[X,X] = \frac{Cov[X,X]}{\sqrt{Var[X]Var[X]}} = \frac{Var[X]}{\sqrt{Var[X]^2}} = 1$$

- Correlation of a random variable X with $-X$ is equal to -1, since:

$$Corr[X,-X] = \frac{Cov[X,-X]}{\sqrt{Var[X]Var[-X]}} = \frac{-Cov[X,X]}{\sqrt{Var[X]Var[X]}} = \frac{-Var[X]}{\sqrt{Var[X]^2}} = -1$$

- For $a,b \in \mathbb{R}$, correlation of a random variable X with $aX+b$ is:

$$Corr[X,aX+b] = \begin{cases} 1, & \text{if } a > 0 \\ -1, & \text{if } a < 0 \\ 0, & \text{if } a = 0 \end{cases}$$

Further Properties of Correlation (cont'd)

- If $Corr[X,Y]$ is close to -1, this means:
  - There is a negative linear dependence relationship between X and Y.
  - Whenever the random variable X takes high values, Y tends to take low values.
  - Whenever the random variable X takes low values, Y tends to take high values.
- **Example:** X can be the price of a second-hand automobile, and Y can be the odometer value.
- If $Corr[X,Y] \approx -1$, this negative linear dependence is very strong.

Further Properties of Correlation (cont'd)

- If $Corr[X,Y]$ is close to 1, this means:
  - There is a positive linear dependence relationship between X and Y.
  - Whenever the random variable X takes high values, Y tends to take high values too.
  - Whenever the random variable X takes low values, Y tends to take low values too.
- **Example:** X can be the grade of a student in PHYS 184 midterm exam, and Y can be the grade of the same student in the final exam.
- If $Corr[X,Y] \approx 1$, this positive linear dependence is very strong.

Further Properties of Correlation (cont'd)

- If $Corr[X,Y] = 0$, this means:
  - There is no linear relationship between X and Y.
  - X and Y are **linearly independent.**
  - They still may be **nonlinearly dependent.**
  - Still there may be some $x, y \in \mathbb{R}$ such that:

$$F_X(x) F_Y(y) \neq F_{XY}(x,y)$$

- **Example:** X can be the number of ships passing through Bosphorus strait on any day and Y can be the number of mobile phones sold in Cevahir Shopping Center on the same day.
- If $Corr[X,Y] \approx 0$, the linear dependence between X and Y is very weak.

Correlation: Example

Q4. Suppose that X and Y have the following joint PMF:

$p_{XY}(x,y)$	y		
x	0	2	4
1	0.20	0.20	0.10
2	0.10	0.05	0.10
3	0.10	0.05	0.10

a. Calculate $Corr[X,Y]$.
b. What type of linear dependence does X and Y have? Is it a strong dependence?

A4. We will compute each component in the following formula:

$$Corr[X,Y] = \frac{Cov[X,Y]}{\sqrt{Var[X]Var[Y]}}$$

Correlation: Example (cont'd)

A4. Remind yourself with the marginal distributions of X and Y.

$$p_X(x) = \begin{cases} 0.50, & x = 1 \\ 0.25, & x = 2 \\ 0.25, & x = 3 \\ 0, & \text{e/w} \end{cases} \qquad p_Y(y) = \begin{cases} 0.40, & y = 0 \\ 0.30, & y = 2 \\ 0.30, & y = 4 \\ 0, & \text{e/w} \end{cases}$$

$$E[X] = 1.75, \qquad E[Y] = 1.8, \qquad Cov[X,Y] = 0.15$$

Correlation: Example (cont'd)

A4. We need to compute $Var[X]$ and $Var[Y]$

$$E\left[X^2\right] = 1^2 \times 0.50 + 2^2 \times 0.25 + 3^2 \times 0.25 = 3.75$$
$$E\left[Y^2\right] = 0^2 \times 0.40 + 2^2 \times 0.30 + 4^2 \times 0.30 = 6$$
$$Var[X] = E\left[X^2\right] - E[X]^2 = 3.75 - (1.75)^2 = 0.6875$$
$$Var[Y] = E\left[Y^2\right] - E[Y]^2 = 6 - (1.8)^2 = 2.76$$

a. As a result:

$$Corr[X,Y] = \frac{Cov[X,Y]}{\sqrt{Var[X]Var[Y]}} = \frac{0.15}{\sqrt{0.6875 \times 2.76}} \approx 0.1089$$

b. There is positive linear dependence between X and Y, though it is not strong.

Correlation: Example II

Q5. Suppose the joint PDF of X and Y is given by:

$$f_{XY}(x,y) = \begin{cases} 2-x-y, & 0 \leq x \leq 1,\ 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find $Corr[X,Y]$ and comment on the relationship between X and Y.

Correlation: Example II

Q5. Suppose the joint PDF of X and Y is given by:

$$f_{XY}(x,y) = \begin{cases} 2-x-y, & 0 \leq x \leq 1,\ 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find $Corr[X,Y]$ and comment on the relationship between X and Y.

A5. Remind yourself with the marginal distributions of X and Y.

$$f_X(x) = \tfrac{3}{2} - x, \quad x \in [0,1], \qquad E[X] = \tfrac{5}{12}$$
$$f_Y(y) = \tfrac{3}{2} - y, \quad y \in [0,1], \qquad E[Y] = \tfrac{5}{12}$$
$$Cov[X,Y] = -\tfrac{1}{144}$$

Correlation: Example II (cont'd)

A5. Now let us compute $Var[X]$ and $Var[Y]$.

$$E\left[X^2\right] = \int_0^1 x^2 f_X(x)\,dx = \int_0^1 \left(\tfrac{3}{2}x^2 - x^3\right)dx = \left[\frac{x^3}{2} - \frac{x^4}{4}\right]_{x=0}^{x=1} = \frac{1}{4}$$

$$Var[X] = E\left[X^2\right] - E[X]^2 = \frac{1}{4} - \left(\frac{5}{12}\right)^2 = \frac{11}{144}$$

Notice that, $Var[Y] = \frac{11}{144}$ since its marginal PDF is the same as the marginal PDF of X. As a result:

$$Corr[X,Y] = \frac{Cov[X,Y]}{\sqrt{Var[X]\,Var[Y]}} = \frac{-\frac{1}{144}}{\sqrt{\frac{11}{144} \times \frac{11}{144}}} = -\frac{1}{11}$$

There is a negative linear dependence between X and Y, though it is not strong.

Variance of the Sum of Two Random Variables

- **Theorem:** Suppose we are given two random variables X and Y. The variance of the sum of these random variables is equal to the sum of their variance plus two times the covariance between these variables. Namely:

$$Var[X+Y] = Var[X] + Var[Y] + 2Cov[X,Y]$$

- **Proof:** For any couple of Random variables X and Y:

$$\begin{aligned}
Var[X+Y] &= E\left[(X+Y)^2\right] - E[X+Y]^2 \\
&= E\left[X^2+Y^2+2XY\right] - (E[X]+E[Y])^2 \\
&= E\left[X^2\right] + E\left[Y^2\right] + 2E[XY] - E[X]^2 - E[Y]^2 - 2E[X]E[Y] \\
&= E\left[X^2\right] - E[X]^2 + E\left[Y^2\right] - E[Y]^2 + 2(E[XY] - E[X]E[Y]) \\
&= Var[X] + Var[Y] + 2Cov[X,Y]
\end{aligned}$$

Variance of the Sum of Two Independent RVs

- Notice that **when X and Y are independent,** $Cov[X,Y]=0$, hence:

$$Var[X+Y]=Var[X]+Var[Y]$$

- In other words, when the variance of the sum of two RVs is equal to the sum of their individual variances, these RVs are independent.

Variance of the Sum of n Random Variables

- **Proposition:** Say X_i, $i = 1, \ldots, n$, is a sequence of random variables. The variance of their sum is equal to the sum of their variances plus two times the covariances of each distinct couple of random variables.

$$
\begin{aligned}
Var\left[X_1 + \ldots + X_n\right] &= Var\left[\sum_{i=1}^{n} X_i\right] \\
&= \sum_{i=1}^{n} Var\left[X_i\right] + \sum_{i=1}^{n} \sum_{\substack{j=1 \\ j \neq i}}^{n} Cov\left[X_i, X_j\right] \\
&= \sum_{i=1}^{n} Var\left[X_i\right] + 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} Cov\left[X_i, X_j\right]
\end{aligned}
$$

Variance of the Sums of Random Variables: Example

Q6. Remind yourself with the following information on the random variables X and Y provided in Question 4:

$$Var[X] = 0.6875, \quad Var[Y] = 2.76, \quad Cov[X,Y] = 0.15$$

a. Calculate $Var[X+Y]$
b. Calculate $Var[X-Y]$
c. Calculate $Var[3X-2Y]$

A6. We will use the following formula:

$$Var[X+Y] = Var[X] + Var[Y] + 2Cov[X,Y]$$

Variance of the Sums of Random Variables: Ex. (cont'd)

a. $Var[X+Y]$ can be computed straightforwardly.

$$\begin{aligned} Var[X+Y] &= Var[X]+Var[Y]+2Cov[X,Y] \\ &= 0.6875+2.76+2\times 0.15 = 3.7475 \end{aligned}$$

b. Notice that $Var[X-Y] = Var[X+(-Y)]$. Hence:

$$\begin{aligned} Var[X-Y] &= Var[X+(-Y)] \\ &= Var[X]+Var[-Y]+2Cov[X,-Y] \\ &= Var[X]+(-1)^2 Var[Y]+2(-1)Cov[X,Y] \\ &= 0.6875+2.76-2\times 0.15 = 3.1475 \end{aligned}$$

Variance of the Sums of Random Variables: Ex. (cont'd)

c. Notice that $Var[3X-2Y] = Var[3X+(-2Y)]$. Hence:

$$\begin{aligned} Var[3X-2Y] &= Var[3X+(-2Y)] \\ &= Var[3X]+Var[-2Y]+2Cov[3X,-2Y] \\ &= 3^2Var[X]+(-2)^2Var[Y]+2(3)(-2)Cov[X,Y] \\ &= 9\times 0.6875+4\times 2.76-12\times 0.15 = 15.4275 \end{aligned}$$

Additive Relationships Between Special Random Variables

- There are several additive relationships between random variables. For example:
  - The sum of random variables of a certain type may create a new type of random variable.
  - **Example:** Sum of independent exponential RVs (all with the same mean) is called an Erlang random variable.
  - **Example II:** Sum of two uniform RVs (both with the same parameter setting) forms a triangular RV.
- We will just examine the most important additive relationships for us.

What Does "iid" Mean?

- Let X and Y be two random variables that are iid.
  - What does the abbreviation "iid" mean?
  - **iid: (i)**ndependent and **(i)**dentically **(d)**istributed.
- The term "iid" implies three things for X and Y.
  1. X and Y are independent.
  $$F_X(x)\,F_Y(y) = F_{XY}(x,y)$$
  2. There is no correlation between X and Y.
  $$Cov[X,Y] = Corr[X,Y] = 0$$
  3. X and Y are identically distributed. They follow the **same distribution** with the **same parameter setting.** Their CDFs are the same.
  $$F_X(x) = F_Y(x) \text{ for all } x \in \mathbb{R}$$

Sum of iid Bernoulli RVs is A Binomial RV

- Let $X_1, \ldots, X_n$, be a sequence of iid Bernoulli RVs.

$$X_i \sim Bern(p) \text{ for all } i = 1, \ldots, n$$

  - X_i are all independent of each other.
  - Each X_i is a Bernoulli RV with success probability p.
- Then:

$$Y = X_1 + X_2 + \ldots + X_n = \sum_{i=1}^{n} X_i$$

is a binomial RV with size parameter n and success probability p.

Sum of iid Bernoulli RVs is A Binomial RV (cont'd)

- Notice that:

$$E[X_i] = p, \quad Var[X_i] = p(1-p), \quad i = 1, \ldots, n$$

Hence[3]:

$$E[Y] = E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{n} p = np$$

$$\begin{aligned} Var[Y] &= Var\left[\sum_{i=1}^{n} X_i\right] \\ &= \sum_{i=1}^{n} Var[X_i] + \sum_{i=1}^{n} \sum_{\substack{j=1 \\ j \neq i}}^{n} Cov[X_i, X_j] \\ &= \sum_{i=1}^{n} p(1-p) = np(1-p) \end{aligned}$$

[3] Since X_i are iid, $Cov[X_i, X_j] = 0$ when $i \neq j$.

Sum of iid Bernoulli RVs is A Binomial RV (cont'd)

- An immediate consequence is as follows.
- Let X_1 be a binomial RV with size parameter n_1 and success probability p, **and independently,** let X_2 be another binomial RV with size parameter n_2 and success probability p.

$$X_1 \sim Binom(n_1, p), \quad X_2 \sim Binom(n_2, p)$$

- Then $Y = X_1 + X_2$ is a binomial RV with size parameter $n_1 + n_2$ and success probability p.

$$Y \sim Binom(n_1 + n_2, p)$$

- That is because Y can be represeneted as a sum of $n_1 + n_2$ iid Bernoulli RVs with success probability p.

Sum of iid Bernoulli RVs is A Binomial RV: Example

Q7. To a party, 6 men and 4 women are invited. Independent of each other, each invited person is likely to participate with probability 0.6. After the party, it is told that 7 people have participated. Then, what is the probability that all 4 women were present at the party?

Sum of iid Bernoulli RVs is A Binomial RV: Example

Q7. To a party, 6 men and 4 women are invited. Independent of each other, each invited person is likely to participate with probability 0.6. After the party, it is told that 7 people have participated. Then, what is the probability that all 4 women were present at the party?

A7. Let X_1 and X_2, respectively, denote the number of men and women that were present. Notice that:

$$X_1 \sim Binom\left(n = 6, p = 0.6\right), \quad X_2 \sim Binom\left(n = 4, p = 0.6\right)$$

and X_1 and X_2 are independent. Notice also:

$$X_1 + X_2 = X \sim Binom\left(n = 10, p = 0.6\right)$$

since X_1 and X_2 has a **common success probability** and they are independent. It is also told that $X_1 + X_2 = 7$ is observed.

Sum of iid Bernoulli RVs is A Binomial RV: Example (cont'd)

A7. The question asks the conditional probability:

$$P\{X_2 = 4 | X_1 + X_2 = 7\}$$

which can expanded as:

$$\begin{aligned} P\{X_2 = 4 | X_1 + X_2 = 7\} &= \frac{P\{X_2=4, X_1+X_2=7\}}{P\{X_1+X_2=7\}} \\ &= \frac{P\{X_2=4, X_1=3\}}{P\{X=7\}} \\ &= \frac{P\{X_2=4\}P\{X_1=3\}}{P\{X=7\}} \end{aligned}$$

Hence:

$$P\{X_2 = 4 | X_1 + X_2 = 7\} = \frac{\binom{4}{4}(0.6)^4(0.4)^0\binom{6}{3}(0.6)^3(0.4)^3}{\binom{10}{7}(0.6)^7(0.4)^3} = \frac{1}{6}$$

Sum of Independent Poisson RVs is A Poisson RV

- Because Poisson RV is a limiting case of Binomial RV, we have the following consequence:
- Let X_1 be a Poisson RV with rate parameter λ_1, **and independently,** let X_2 be another Poisson RV with rate parameter λ_2.

$$X_1 \sim Pois\left(\lambda_1\right), \quad X_2 \sim Pois\left(\lambda_2\right)$$

- Then $Y = X_1 + X_2$ is a Poisson RV with rate parameter $\lambda = \lambda_1 + \lambda_2$.

$$Y \sim Pois\left(\lambda_1 + \lambda_2\right)$$

Sum of Independent Poisson RVs is A Poisson RV (cont'd)

- More generally, let X_i be a sequence of independent Poisson RVs with respective rate parameters λ_i.

$$X_i \sim Pois\left(\lambda_i\right) \text{ for all } i = 1, \ldots, n$$

Then:

$$Y = X_1 + X_2 + \ldots + X_n = \sum_{i=1}^{n} X_i$$

is also a Poisson random variable with rate parameter:

$$\lambda = \lambda_1 + \lambda_2 + \ldots + \lambda_n = \sum_{i=1}^{n} \lambda_i$$

namely

$$Y \sim Pois\left(\lambda = \sum_{i=1}^{n} \lambda_i\right)$$

Sum of Independent Poisson RVs is A Poisson RV (cont'd)

- Notice that:

$$E[X_i] = \lambda_i, \quad Var[X_i] = \lambda_i, \quad i = 1, \ldots, n$$

Hence[4]:

$$E[Y] = E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{n} \lambda_i$$

$$Var[Y] = Var\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} Var[X_i] + \underbrace{\sum_{i=1}^{n} \sum_{\substack{j=1 \\ j \neq i}}^{n} Cov[X_i, X_j]}_{0} = \sum_{i=1}^{n} \lambda_i$$

[4] Since X_i are iid, $Cov[X_i, X_j] = 0$ when $i \neq j$.

Sum of Independent Poisson RVs is A Poisson RV: Ex.

Q8. To a bus stop, average arrival rate of male passengers is 3 passengers per minute and average arrival rate of female passengers is 5 passengers per minute. If the arrivals are independent and follow Poisson distribution, what is the probability that at most 3 passengers arrive in the following minute?

A8. Let X_1 and X_2 denote the number of arrivals, respectively of male and female passengers.

$$X_1 \sim Pois\,(\lambda_1 = 3) \text{ and } X_2 \sim Pois\,(\lambda_2 = 5)$$

The question states that X_1 and X_2 are independent. Hence $X = X_1 + X_2$ is also a Poisson RV with $\lambda = \lambda_1 + \lambda_2 = 8$.

$$X_1 + X_2 = X \sim Pois\,(\lambda = \lambda_1 + \lambda_2 = 8)$$

Sum of Ind. Poisson RVs is A Poisson RV: Ex. (cont'd)

A8. The question asks for $P\{X \leq 3\}$.

$$\begin{aligned} P\{X \leq 3\} &= \sum_{x=0}^{3} P\{X = i\} \\ &= \sum_{x=0}^{3} e^{-8} \frac{8^x}{x!} \\ &= e^{-8} \left(\frac{8^0}{0!} + \frac{8^1}{1!} + \frac{8^2}{2!} + \frac{8^3}{3!} \right) \approx 0.0424 \end{aligned}$$

Sum of iid Geometric RVs is A Negative Binomial RV

- Let X_i be a sequence of iid Geometric RVs with success probability p.

$$X_i \sim Geom(p) \text{ for all } i = 1, \ldots, r$$

Then:

$$Y = X_1 + X_2 + \ldots + X_r = \sum_{i=1}^{r} X_i$$

is a negative binomial random variable with number of necessary successes r and success probability p:

$$Y \sim NBinom(r, p)$$

Sum of iid Geometric RVs is A Neg. Binomial RV (cont'd)

- Notice that:

$$E[X_i] = \frac{1}{p}, \quad Var[X_i] = \frac{1-p}{p^2}, \quad i = 1, \ldots, r$$

Hence[5]:

$$E[Y] = E\left[\sum_{i=1}^{r} X_i\right] = \sum_{i=1}^{r} E[X_i] = \sum_{i=1}^{r} \frac{1}{p} = \frac{r}{p}$$

$$\begin{aligned} Var[Y] &= Var\left[\sum_{i=1}^{r} X_i\right] \\ &= \sum_{i=1}^{r} Var[X_i] + \sum_{i=1}^{r} \sum_{\substack{j=1 \\ j \neq i}}^{r} Cov[X_i, X_j] \\ &= \sum_{i=1}^{r} \frac{1-p}{p^2} = \frac{r(1-p)}{p^2} \end{aligned}$$

[5] Since X_i are iid, $Cov[X_i, X_j] = 0$ when $i \neq j$.

Sum of iid Geometric RVs is A Neg. Binomial RV: Example

Q9. In a basketball game, James is likely to score a 3-pts shot with probability 0.3. We assume independence between the success of his each shot.

The narrator of the game tells us that James has scored his second 3-pts in his eighth shot. What is the probability that his first 3-pts score was in his third shot?

Sum of iid Geometric RVs is A Neg. Binomial RV: Example

Q9. In a basketball game, James is likely to score a 3-pts shot with probability 0.3. We assume independence between the success of his each shot.

The narrator of the game tells us that James has scored his second 3-pts in his eighth shot. What is the probability that his first 3-pts score was in his third shot?

A9. Let X_1 denote the number of shots that James need to take until his first 3-pts score. After his first 3-points score, let X_2 denote the number of **additional shots** that James need to take until his second 3-pts score. X_1 and X_2 are iid geometric RVs.

$$X_1 \sim Geom\left(p = 0.3\right), \quad X_2 \sim Geom\left(p = 0.3\right)$$

Sum of iid Geom. RVs is A Neg. Binom. RV: Ex. (cont'd)

A9. We define $X = X_1 + X_2$ as the total number of shots that James need to take until his second 3-pts score. X is a negative binomial RV:

$$X \sim NBinom\,(r = 2, p = 0.3)$$

According to what narrator says, $X = X_1 + X_2 = 8$ is observed. The question, then, asks the following conditional probability.

$$P\{X_1 = 3 | X = 8\}$$

Sum of iid Geom. RVs is A Neg. Binom. RV: Ex. (cont'd)

A9. The conditional probability can expanded as:

$$\begin{aligned} P\{X_1 = 3 | X_1 + X_2 = 8\} &= \frac{P\{X_1=3, X_1+X_2=8\}}{P\{X_1+X_2=8\}} \\ &= \frac{P\{X_1=3, X_2=5\}}{P\{X=8\}} \\ &= \frac{P\{X_1=3\}P\{X_2=5\}}{P\{X=8\}} \end{aligned}$$

since X_1 and X_2 are independent. Hence:

$$P\{X_1 = 3 | X = 8\} = \frac{\left(p(1-p)^2\right)\left(p(1-p)^4\right)}{\binom{7}{1} p^2 (1-p)^6} = \frac{1}{7}$$

In fact, if James second 3-pts score is in his eighth shot, then the first 3-pts score is equally likely to be in any of his first seven shots.

Sum of Normal RVs is A Normal RV

- Let X_1 be a normal RV with mean μ_1 and standard deviation σ_1, **and independently,** let X_2 be another normal RV with mean μ_2 and standard deviation σ_2.

$$X_1 \sim N(\mu_1, \sigma_1), \quad X_2 \sim N(\mu_2, \sigma_2)$$

- Then $Y = X_1 + X_2$ is a normal RV with mean:

$$\mu = \mu_1 + \mu_2$$

and standard deviation:

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2}$$

Namely:

$$Y \sim N\left(\mu = \mu_1 + \mu_2, \sigma = \sqrt{\sigma_1^2 + \sigma_2^2}\right)$$

Sum of Normal RVs is A Normal RV (cont'd)

- Now suppose, X_1 and X_2 are not independent, and their correlation is ρ.

$$X_1 \sim N(\mu_1, \sigma_1), \quad X_2 \sim N(\mu_2, \sigma_2), \quad Corr[X_1, X_2] = \rho$$

- Then $Y = X_1 + X_2$ is a normal RV with mean:

$$\mu = \mu_1 + \mu_2$$

and standard deviation

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2}$$

Namely:

$$Y \sim N\left(\mu = \mu_1 + \mu_2, \sigma = \sqrt{\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2}\right)$$

Sum of Normal RVs is A Normal RV: Example

Q10. Let X denote the amount of rainfall to İstanbul in milimeters on a randomly selected day of the year and let Y denote the same for Trabzon on the same day. It is known that:

- X is normally distributed with mean $\mu_X = 1.2\ mm$ and variance $\sigma_X^2 = 0.09\ mm^2$,
- Y is normally distributed with mean $\mu_Y = 1.6\ mm$ and variance $\sigma_Y^2 = 0.04\ mm^2$,
- the correlation between X and Y is $\rho_{XY} = 0.4$.

What is the probability that the **total amount** of rainfall to Istanbul and Trabzon on any day to be greater than 3.5 mm?

A10. The question provides:

$$X \sim N(\mu_X = 1.2, \sigma_X = 0.3), \quad Y \sim N(\mu_Y = 1.6, \sigma_Y = 0.2)$$

and asks the probability $P\{X + Y \geq 3.5\}$.

Sum of Normal RVs is A Normal RV: Example (cont'd)

A10. Note that $X+Y$ is also normally distributed with mean:

$$\mu = \mu_X + \mu_Y = 2.8$$

and standard deviation:

$$\begin{aligned}\sigma &= \sqrt{\sigma_X^2 + \sigma_Y^2 + 2\rho\sigma_X\sigma_Y} \\ &= \sqrt{0.09 + 0.04 + 2(0.4)(0.3)(0.2)} = \sqrt{0.178} \\ &\approx 0.422\end{aligned}$$

namely:

$$X+Y \sim N(\mu = 2.8, \sigma = 0.422)$$

The requested probability is found as:

$$\begin{aligned}P\{X+Y \geq 3.5\} &= P\left\{Z \geq \tfrac{3.5-2.8}{0.422}\right\} \approx P\{Z \geq 1.659\} \\ &= 1 - \Phi(1.659) \approx 0.0486\end{aligned}$$

Sum of Normal RVs is A Normal RV: Example II

Q11. Let X and Y be defined as:

$$X \sim N(\mu_X = 10, \sigma_X = 2), \quad Y \sim N(\mu_Y = 15, \sigma_Y = 3), \quad \rho_{XY} = -0.5$$

Compute $P\{3X - 2Y \leq 10\}$.

Sum of Normal RVs is A Normal RV: Example II

Q11. Let X and Y be defined as:

$$X \sim N(\mu_X = 10, \sigma_X = 2), \quad Y \sim N(\mu_Y = 15, \sigma_Y = 3), \quad \rho_{XY} = -0.5$$

Compute $P\{3X - 2Y \leq 10\}$.

A11. We know that $3X - 2Y$ is also normally distributed. Let us compute its mean and variance:

$$E[3X - 2Y] = 3E[X] - 2E[Y] = 3 \times 10 - 2 \times 15 = 0$$

$$\begin{aligned} Cov[X,Y] &= Corr[X,Y]\sqrt{Var[X]Var[Y]} \\ &= \rho_{XY}\sigma_X\sigma_Y = (-0.5)(2)(3) = -3 \end{aligned}$$

$$\begin{aligned} Var[3X - 2Y] &= Var[3X] + Var[-2Y] + 2Cov[3X, -2Y] \\ &= (3)^2 Var[X] + (-2)^2 Var[Y] + 2(3)(-2)Cov[X,Y] \\ &= 9 \times 4 + 4 \times 9 - 12 \times (-3) = 108 \end{aligned}$$

Sum of Normal RVs is A Normal RV: Example II (cont'd)

A11. Hence:

$$3X - 2Y \sim N\left(\mu = 0, \sigma = \sqrt{108}\right)$$

The desired probability can be computed as follows:

$$\begin{aligned} P\{3X - 2Y \leq 10\} &= P\left\{Z \leq \frac{10-0}{\sqrt{108}}\right\} \approx P\{Z \leq 0.962\} \\ &= \Phi(0.962) \approx 0.8320 \end{aligned}$$

Sum of Normal RVs is A Normal RV (cont'd)

- More generally, let X_i be a sequence of normal RVs with respective means μ_i and standard deviations σ_i.

$$X_i \sim N(\mu_i, \sigma_i) \text{ for all } i = 1, \ldots, n$$

- Suppose further, each couple of RV in this sequence has a certain correlation (they are not independent).

$$Corr[X_i, X_j] = \rho_{ij} \text{ for all } i = 1, \ldots, n,\ j = 1, \ldots, n, \text{ and } i \neq j$$

Sum of Normal RVs is A Normal RV (cont'd)

- Then:

$$Y = X_1 + X_2 + \ldots + X_n = \sum_{i=1}^{n} X_i$$

is also a normal random variable with mean:

$$\mu = \mu_1 + \mu_2 + \ldots + \mu_n = \sum_{i=1}^{n} \mu_i$$

and standard deviation:

$$\sigma = \sqrt{\sum_{i=1}^{n} \sigma_i^2 + 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \rho_{ij} \sigma_i \sigma_j}$$

END OF LECTURE SLIDE 10